Single-LED 3D Visible Light Positioning with a Reorientable Photodiode

Experimental results on the Cambridge testbed — executive summary

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Highlights

- **3.7 cm mean 3D error with a single LED and a single photodiode.** Over 125 positions on 5 heights (median 3.4 cm, P90 6.1 cm, no failures), with $K = 10$ orientations. With $K = 5$ the error is the same (3.7 cm); with $K = 3$ it is 4.9 cm.
- **Validated live.** The real-time demo on 131 random, blind positions ($K = 5$) gives 4.1 cm (median 3.9 cm).
- **Calibration needs only 9 positions.** A 3×3 grid over the working area, with a flat PD, gives 3.5 cm, the same as the 19 881-point map. It also yields the LED XY position. *Where* the points are matters more than *how many*: 9 points clustered around the LED give 17–22 cm.
- **The positioning system diagnosed its own testbed.** From the RSS alone, VLP revealed that the gantry Z axis travels only $\approx 0.63\times$ the commanded distance. A tape measurement later gave $0.6\times$. Correcting it halves the error (7.1 → 3.7 cm).
- **The 700 kHz LED is tilted by $\approx 5°$.** Its radiation pattern is centred 18 cm away from its physical position, so the method is a diagnostic tool for the installation too.
- **The photodiode angular model is decisive.** The SQ response (Bastiaens 2020) gives 3.9 cm; the usual $\cos^{m_R}$ and Lambertian cos models give 8.4 cm and 77 cm.
- **Two LEDs give position and yaw without any power calibration.** Intersecting the directions to two LEDs, the minimum for crossing rays (e.g. a corridor), needs no LED gain and no Lambertian order. With the best pair (500 + 900 kHz) the error is 3.3 cm with known yaw, and 3.6 cm with the yaw estimated too (1.2° yaw error). This is a path to IMU-free attitude.

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1. Setup in brief

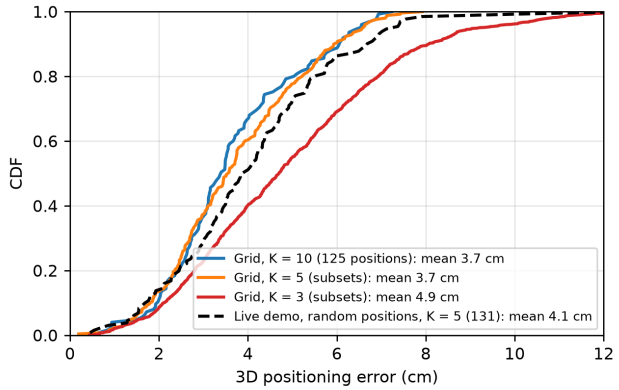
- **Single-LED system.** The system under study uses **one LED** on the ceiling (3.0 m, facing down, modulated at 500 kHz). All results refer to this LED unless stated otherwise.
- **Receiver.** A Thorlabs PDA100A2 photodiode on a 15° adapter, turned by a motorized rotation stage to K azimuths.
- **Additional LEDs, for comparison only.** The testbed has three more LEDs (300, 700 and 900 kHz). Their RSS is recorded in the same measurements by frequency separation. They are used only in the comparisons with two or more LEDs (F6) and for the installation diagnosis (F5).
- **Ground truth.** An igus XYZ gantry: 1.4×1.4 m, PD height 0.95–1.32 m.
- **RSS measurement.** Lock-in on 100 ms raw records (1 Mpts, Keysight MSO-X 4154A).
- **Estimator.** The direction comes from the *shape* of the K RSS (NLS, SQ PD response). The range is in closed form from the same RSS, with the calibrated LED gain A and Lambertian order m.
- **Datasets.**
  - 125 positions $\times$ $K = 10$ (1 250 measurements, regular 5×5 grid $\times$ 5 heights);
  - a flat-PD RSS map of 19 881 points (10 mm grid);
  - 131 random positions from the live demo;
  - the link frequency response (5 kHz–1 MHz).

- **Tooling.** The acquisition tool and the demo run unattended for many hours. Raw-socket SCPI over LAN replaced the vendor VISA, whose large transfers froze the oscilloscope: 1 500 consecutive 1-Mpt captures ran without error. Interrupted campaigns recover automatically and can be resumed.

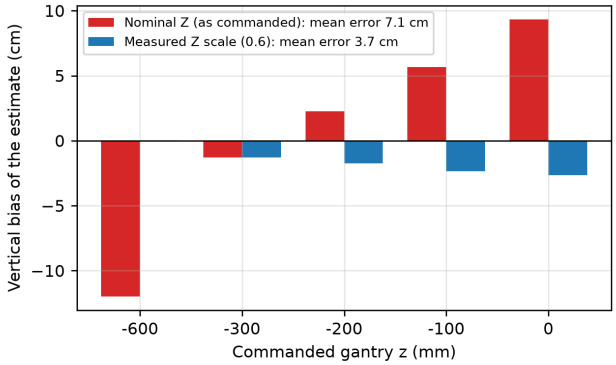
2. Main results

Configuration (500 kHz LED unless stated)	Positions	Mean	Median	P90
Single LED, $K = 10$, regular 3D grid	125	**3.66**	3.40	6.05
$K = 5$ (all subsets of the 10 angles)	125	3.73	3.56	5.94
$K = 3$	125	4.94	4.70	8.07
Calibrated with only 9 points (3×3), LED XY known	125	3.54	3.17	6.48
Calibrated with only 9 points, LED XY also calibrated	125	3.47	3.13	6.35
Live demo, random positions, $K = 5$	131	4.10	3.91	6.68
Two LEDs (500 + 900 kHz), directions only (no A, m)	50	**3.30**	2.50	6.27
same, yaw unknown and estimated	50	3.60	3.06	7.03
Four LEDs, directions only	50	2.77	2.46	5.34
same, yaw unknown and estimated	50	2.92	2.50	5.33
What happens if a model element is wrong (mean error, same 125 positions):				
Gantry Z taken as commanded (no 0.6 scale)	7.1 cm (vertical bias from −12 to +9 cm)			
PD response $\cos^{1.9}\psi$ / $\cos\psi$ instead of SQ	8.4 cm / 77 cm			
Calibration points clustered around the LED (3×3, 0.2 m)	17.3 cm			

Table 1: 3D positioning error in cm. The multi-LED rows use the 50 positions of the first campaign (on those positions the single-LED method gives 2.98 cm), with the positions of the other LEDs obtained from their ray intersections at the other height.



(a) Error CDF: regular grid ($K = 10$, 5, 3) and live demo.



(b) Vertical bias per gantry height: nominal Z vs. measured scale.

Figure 1: Accuracy and the gantry Z-axis diagnosis.

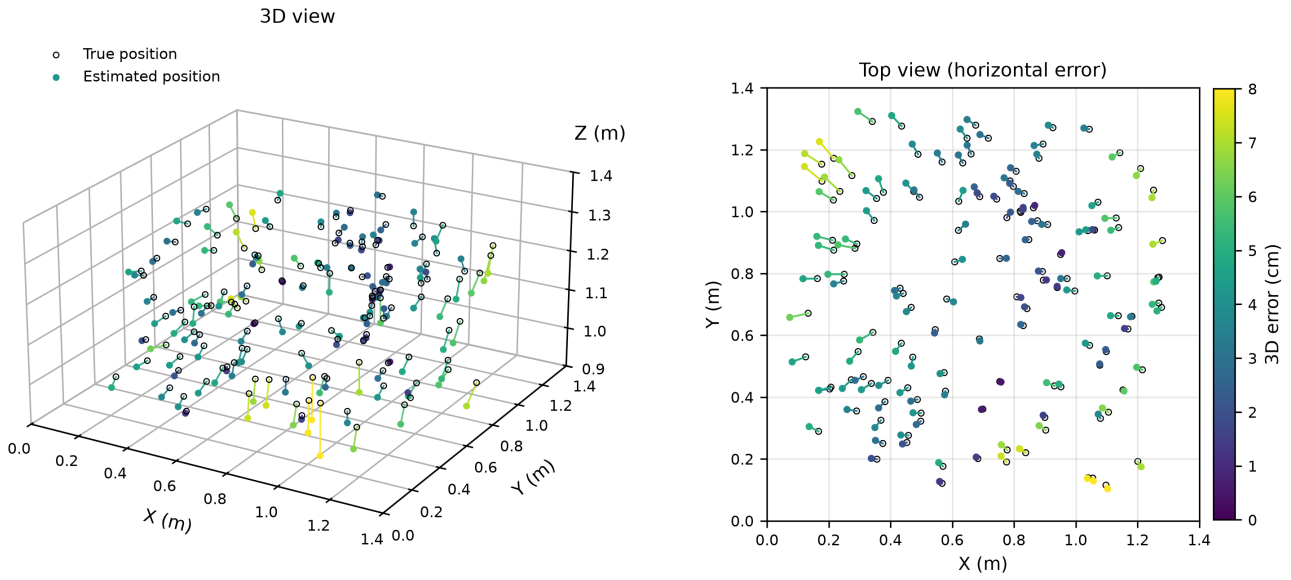


Figure 2: Live demo: 131 random positions, single LED, $K = 5$. True positions (hollow) and estimates (coloured by 3D error), joined by their error vectors.

3. Key findings

F1. Centimetre-level 3D positioning from one LED and one PD (Fig. 2).

- Direction and range come from the same K RSS. No extra cooperative measurement is needed.
- $K = 5$ performs like $K = 10$, and $K = 3$ still works without failures, but with no redundancy.
- The remaining error is dominated by model bias, not by noise: more orientations do not reduce it.

F2. The PD angular response must be modelled correctly. The shape test of the K RSS selects the SQ model independently of A and m. Lambertian receiver models fail by a factor of 2 to 20.

F3. Self-calibration of the testbed geometry. With the nominal Z axis, the vertical bias grew linearly with height, and A, m and the adapter tilt seemed to depend on the PD height.

- Two independent principles (amplitude and ray triangulation) both gave a Z scale of 0.63–0.65.
- A tape measurement then confirmed 0.6.
- With it, A stays constant within $\pm 0.2\,\%$ and the adapter tilt within $\pm 0.02°$ across heights.

F4. Calibration budget: 9 well-placed points (Fig. 3). Only A and m are unknown.

- One point below the LED fixes A but not m.
- Two points (nadir + one at $\approx 20°$) are the theoretical minimum, but fragile.
- A regular 3×3 or 4×4 grid over the working area reaches the full-map result (3.5–3.6 cm). Ten random points give 3.8 cm; points clustered near the LED fail.
- The reason: the measured pattern is *not* a single $\cos^m$. The apparent m drops from ≈ 11 near the axis to ≈ 6 beyond 20°. The calibration points must therefore cover the angles used for positioning. The $\cos^m$ model floor is 3.1 cm.
- The same 9 points also calibrate the LED XY (within 3 mm, 3.5 cm error). The LED *height* cannot be identified from a single plane: it must be measured, or a second plane is needed.

F5. Installation diagnosis of the other LEDs (Fig. 4). The 9-point calibration gives the same result as the full map for every LED, but the 300, 900 and 700 kHz LEDs position worse (6–17 cm) because they are not pointing straight down.

- Their pattern centres are shifted by 2.6, 6 and 18 cm from their physical positions, i.e. tilts of $\approx 0.8°$, 1.9° and 5.4°.
- The calibration of a flat-PD plane alone cannot separate an LED tilt from an LED offset.

F6. Multi-LED, directions only. Intersecting the directions to several LEDs needs neither the LED power nor A and m.

- **Two LEDs** are enough, the realistic case of a corridor, and the minimum to cross rays and to estimate the yaw. With the best pair (500 + 900 kHz): 3.3 cm with known yaw, 3.6 cm estimating the yaw too (1.2° yaw error).
- With the four LEDs: 2.8 cm and 2.9 cm (0.7° yaw error).
- The pairs that include the tilted 700 kHz LED are the worst (6.3 cm), consistent with F5.

F7. Acquisition time is dominated by the rotation stage.

- 3.7 min per position with $K = 5$ (live demo average) and 4.2 min with $K = 10$.
- About 86 % of that time is the MTRS rotation at 1.5°/s; the measurements themselves take a few seconds.
- By estimate, a rotation stage of $\sim 20°/s$ would bring a $K = 5$ position below one minute: faster demos, larger datasets and moving receivers.

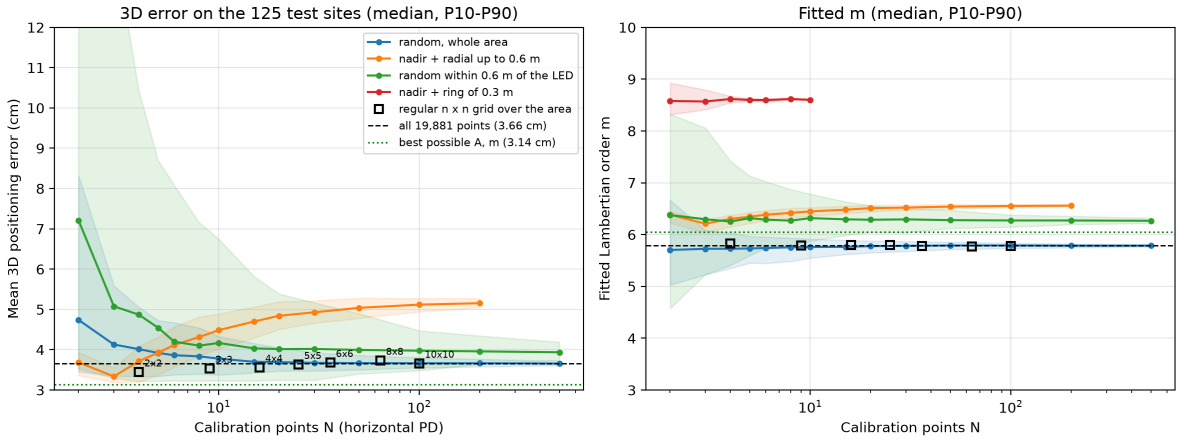


Figure 3: Calibration budget (500 kHz LED, flat PD): 3D error on the 125 positions vs. number of calibration points for several layouts (left) and the fitted m (right). Squares: regular $n \times n$ grids over the working area.

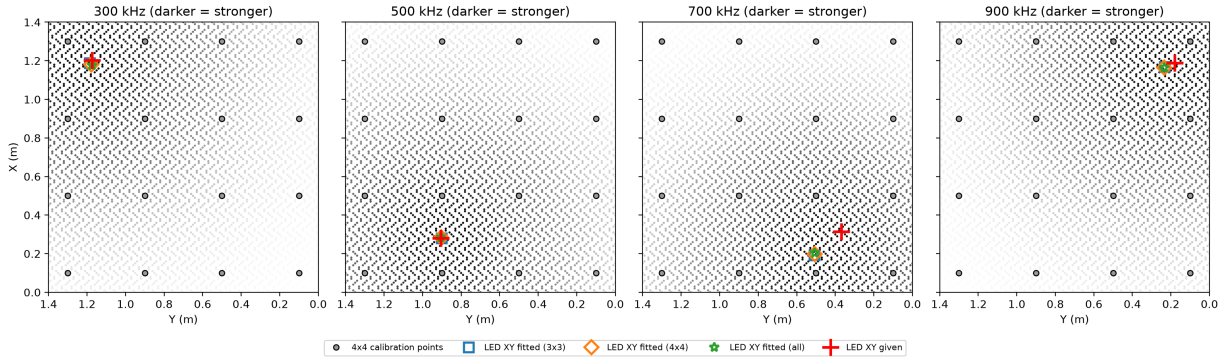


Figure 4: Flat-PD RSS map of each LED. Markers: the LED XY fitted from the calibration points (centre of the radiation pattern) and the physical LED position (red cross). The 700 kHz LED shows an 18 cm offset, i.e. a $\approx 5°$ tilt.

4. Most promising directions

1. **Journal paper on the experimental validation** (immediate).
   - A dense regular 3D grid plus a blind random test set, with a strict calibration/test split.
   - Comparison with the position error bound (PEB), using the noise measured from repeated records.
   - The calibration budget and the self-diagnosis results as methodological contributions.
2. **A better LED pattern model** to break the 3.1 cm floor of $\cos^m$: an angle-dependent m, a spline, or a physics-informed NN. Calibrating on *two* planes would also identify the LED height and tilt.
3. **Two-LED / multi-LED positioning with yaw estimation**: towards IMU-free 3D position *and* attitude for mobile receivers (corridors, drones). Next step: fusing directions and ranges.

4. **Robustness studies.**
   - Arbitrary but known orientations (no fixed codebook).
   - Orientation errors (an IMU instead of encoders).
   - Occlusion of some orientations: detect them (e.g. with the residual of the RSS-shape fit) and estimate with the remaining ones. The results above with $K = 5$ and $K = 3$ suggest that a reduced set of orientations can still position.
5. **Faster acquisition.** A faster rotation stage (F7), continuous rotation with on-the-fly sampling (hundreds of orientations in the time of $K = 10$), or a pyramid angle-diversity receiver for snapshot positioning.
6. **Data for machine learning.** A dense, angle-major dataset (one MTRS angle per scan) could provide $\sim$30 000 labelled measurements in one day, for a hybrid model-based/learning collaboration.

5. Equipment requests

- **Faster motorized rotation stage** ($\gtrsim 20°/s$): the current stage is the bottleneck of every experiment (F7).
- **Automatic laser distance meter**: to measure the PD and LED heights and positions quickly and reproducibly. Today the ground truth of the heights relies on a tape (F3, F4).
- **IMU** on the receiver: orientation measured with realistic errors, needed for the robustness study and the step towards drones.

6. Points to discuss

1. **Journal timing.** Are the current results, together with the theory (receiver-side reorientation, PEB), enough to start writing the journal paper? Or should the writing wait for the dense 3D grid, the blind test set and the PEB comparison?
2. **Scope and venue.** Single LED only, or also include the two-LED / multi-LED results? Target journal (e.g. IEEE TIM, IEEE Sensors Journal or IEEE IoT Journal)?
3. **Equipment.** Priority and budget of the rotation stage, the laser distance meter and the IMU.
4. **LED installation.** Realign the tilted LEDs (700 kHz), or keep them and model the LED tilt (a more general calibration, but not comparable with the data already recorded)?
5. **Collaborations.** Scope of a possible collaboration on machine learning and of data sharing.